

1. **How is the hydropower situation in Nepal? Describe briefly. Make sketch of run-off river project and show its all components.**

Answer:

Nepal has very high hydropower potential because of its steep Himalayan topography, perennial rivers and large elevation differences. The country's major rivers: Koshi, Gandaki, Karnali provide excellent opportunities for hydropower development. Nepal is also considered as the second richest country in the world in terms of water resources. There are more than 6000 big and small rivers in Nepal with the estimated economically feasible hydropower potential of about 42,000 MW. By mid-2026, total installed electricity generation capacity reached about 4,120–4,236 MW.

Most of the hydropower projects in Nepal are run-of-river (RoR) schemes because Nepal has steep river gradients and suitable topography. River discharge is very high during the monsoon but considerably lower in the dry season. Therefore, electricity generation from RoR projects also varies seasonally. Hence, storage projects are also essential. Storage projects such as Kulekhani HPP has helped regulate water availability and provide electricity during periods of low river flow. The Department of Electricity Development lists Kulekhani-I (60 MW) and Kulekhani-II (32 MW) among Nepal's operating hydropower plants.

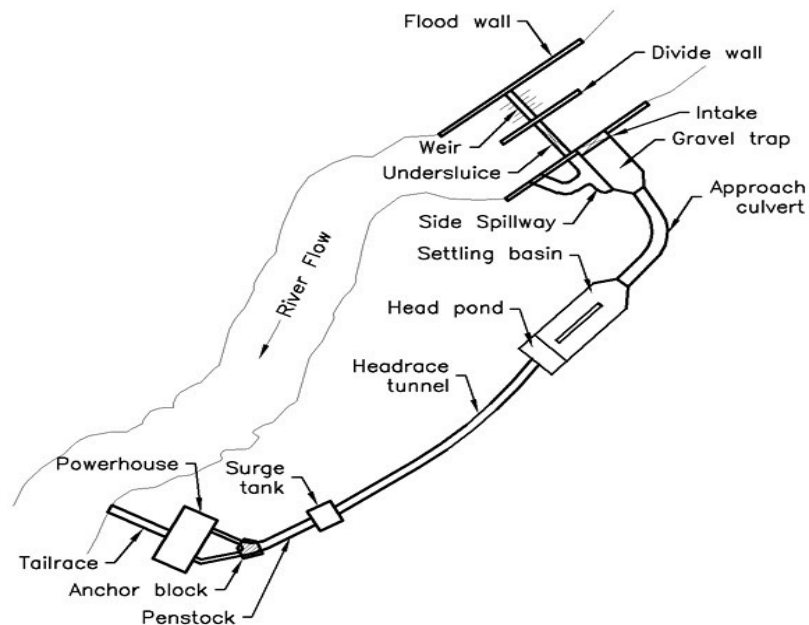
Numerous projects are currently operating, under construction, or being licensed. For example, the Department of Electricity Development continues to list many projects exceeding 1 MW under survey and construction licenses. Nepal has increasingly developed hydropower not only for domestic consumption but also for electricity export to neighboring countries, particularly India. Nepal also officially began its historic electricity export to Bangladesh via India's grid on November 15, 2024. Hence the country has considerable scope for further development of hydropower for domestic industrialization, electrification and regional electricity trade.

However, recent flash flood at Tiber-Nepal border, Rasuwa and Nuwakot that struck on 26th August, 2026 caused several hydropower projects to be severely damaged where hundreds of workers and engineers have gone missing inside its tunnel. Some of the hydropower projects that was severely affected by flash floods are Chilime Hydropower Plant (22 MW), Upper Trishuli-1 (216 MW) & Rasuwa-Bhotekoshi (120 MW), Rasuwagadhi Hydropower Project (111 MW) and so on.

Run-of-river plants:

- RoR hydropower plants do not regulate the hydrograph of the river by one or more seasons.
- The plant utilizes the minimum flow in a river having no appreciable pondage at its upstream.
- In this type of plants, weir or barrage construct across a river, simply to rise and maintain the water level at certain height.
- Usually, low head (up to 70 m) scheme and is suitable only on a perennial river with sufficient dry weather flow to generate designed power.
- Limited storage is used to meet hourly or weekly function of flow.
- Eg: Khimti HPP, Bhotekoshi HPP etc.

The layout and major components of a typical run-of-river hydropower plant are given below:



Components:

i. Weir:

- It is constructed across the river to divert the part of river flow to various water conveyance systems through the intake.

ii. Intake:

- It is a structure constricted to divert the required amount of water into a conduit leading to the power plant.
- Trash racks are provided to prevent entry of floating debris and coarse bed load into water conveyance systems.

iii. Gravel trap:

- It is required to flush out bed sediments that enter the approach canal back into a river.
- The gravel trap is constructed close to the intake in order to prevent gravel from getting into the approach channel.

iv. Settling basin:

- It is the structure to remove suspended sediments from the conveyance water for power plant.
- It shall be designed to ensure that the water entering the water conveyance system is free from sediments that can damage the penstock and turbine runners.

v. Headrace conveyance:

- A canal or pipe or tunnel or combination of these are provided for delivering the design flow to the forebay or surge tank with a minimum energy loss.

vi. Forebay:

- It is located at the beginning of penstock shaft that supplies required flow to the turbine during start up, accommodate the reject flow during shut down, reduce water hammer effect.
- It serves as a secondary settling basin as well as a sediment sluice to remove settle suspended particles.

vii. Surge tank:

- It is located between the headrace pressure conduits and the steeply sloping penstock pipe.
- It is generally a cylindrical storage reservoir connected to penstock pipe close to turbine.

viii. Penstock:

- It is a pipe that conveys the flow from forebay or surge tank to the turbine.
- They are designed to carry water to turbines with the least possible loss of head consistent with the overall economy of the project.

ix. Anchor blocks:

- It is an encasement of a penstock designed to restrain the pipe movement in all durations.
- Anchor block should be placed at all sharp horizontal and vertical bends, as there are forces at such bends which tends to move the pipe out of the designated alignment.

x. Powerhouse:

- It is a structure complex where all the equipment for providing electricity is suitably arranged.
- Two requirements of powerhouse planning are functional efficiency and aesthetic beauty.

xi. Tailrace:

- It disposes the design flow discharged by hydraulic turbine after the power generation.

2. Elaborate the importance of reservoir project in context of Nepal.

Answer:

A reservoir project is a water-resources project in which water is stored behind a dam and released according to demand. In Nepal, where river flows vary greatly between the monsoon and dry seasons, reservoir projects are particularly important for effective utilization of water resources.

i. Hydropower Generation

- Nepal has enormous hydropower potential, but river discharge is highly seasonal.
- Reservoirs store excess monsoon water and release it during the dry season.
- Reservoir hydropower can also meet peak electricity demand as generation can be adjusted according to demand.

ii. Flood Control

- Reservoirs can temporarily store a portion of excessive monsoon runoff.
- Controlled release reduces downstream flood peaks and helps protect settlements, agricultural land, roads and infrastructure.
- This is particularly important in the Terai and other flood-prone areas.

iii. Irrigation Development

- Agriculture in Nepal depends heavily on monsoon rainfall.
- Stored water can be supplied during dry periods, allowing year-round irrigation.
- It can increase cropping intensity and agricultural productivity.

iv. Drinking Water Supply

- Reservoirs can provide a reliable source of water for urban and rural drinking-water systems.
- Stored water can help meet demand during prolonged dry periods.

v. Regulation of River Flow

- Reservoirs regulate highly variable river discharge.
- They can maintain a more uniform downstream flow during the dry season, benefiting irrigation, water supply and other water-dependent activities.

vi. Multipurpose Water Resource Development

A single reservoir can serve several purposes simultaneously, such as:

- Hydropower generation
- Irrigation
- Flood control
- Drinking-water supply
- Industrial water supply
- Recreation and tourism
- Fisheries

vii. **Economic Development**

- Large reservoir projects create employment during construction and operation.
- Reliable electricity and irrigation support industries, agriculture, trade and tourism.
- Hydropower generation can also increase government revenue and export potential.

3. **Briefly explain the safety measures needed to be considered inside the underground power house.**

Answer:

An underground powerhouse is a confined working environment where risks arise from fire, electrical equipment, ventilation, water leakage, falling rocks, machinery, and limited access.

The recent catastrophic flood in Rasuwa clearly demonstrates why strict safety rules and disaster preparedness are essential in hydropower construction. The sudden flooding severely affected hydropower projects and infrastructure along the Bhotekoshi–Trishuli river corridor. The mud water flooded the entire tunnel where hundreds of workers have gone missing.

This disaster highlighted the serious risks faced by workers at hydropower construction sites. Rescue operations focused on workers affected or trapped at project sites and tunnels are underway. Hence, the reliable emergency communication, evacuation procedures, rescue access, and emergency preparedness is very essential.

The following safety measures should be considered inside the underground power house:

- i. **Adequate ventilation**
 - a. Provide mechanical ventilation to supply fresh air and remove heat, smoke, dust and harmful gases.
 - b. Monitor oxygen and toxic-gas levels where necessary.
- ii. **Fire protection**
 - a. Install fire alarms, smoke detectors, fire extinguishers and suitable fire-fighting systems.
 - b. Provide fire-resistant doors and clearly marked emergency exits.
 - c. Maintain separate and safe storage of flammable materials.
- iii. **Electrical safety**
 - a. Properly earth all electrical equipment.
 - b. Provide insulation, circuit breakers, protective relays and emergency shutdown systems.
 - c. Restrict access to high-voltage areas to authorized personnel only.
- iv. **Protection against water**
 - a. Provide effective drainage, sump pumps and standby pumping arrangements.
 - b. Monitor seepage and water pressure, particularly around the powerhouse cavern and tunnels.
- v. **Rock and structural stability**
 - a. Ensure proper rock bolting, shotcrete, lining and other support systems.
 - b. Regularly inspect the roof, walls and support structures for cracks, loose rocks and deformation.
- vi. **Safe access and escape**
 - a. Provide properly illuminated stairways, walkways and emergency exits.
 - b. Keep escape routes free from obstruction and clearly mark them with signs.
- vii. **Lighting and communication**

- a. Provide adequate normal and emergency lighting.
 - b. Install reliable telephone/radio communication with the control room and outside emergency services.
 - viii. **Machinery and moving equipment safety**
 - a. Guard rotating and moving parts of turbines, generators and auxiliary machinery.
 - b. Provide emergency-stop devices and maintain safe working distances.
 - ix. **Personal protective equipment (PPE)**
 - a. Workers should use helmets, safety shoes, gloves, eye/ear protection and other PPE appropriate to the work.
 - x. **Emergency preparedness**
 - a. Prepare emergency procedures for fire, flooding, electrical accidents, rockfall and equipment failure.
 - b. Conduct regular safety drills and ensure workers know evacuation procedures.
4. **Classify the institutions involved in hydropower development in Nepal and explain their roles.**

Answer:

Hydropower development in Nepal involves government ministries, regulatory agencies, public utilities, investment institutions, private developers, and supporting organizations. Their roles can be broadly classified as follows.

i. **Policy-making and Government Institutions**

a. Ministry of Energy, Water Resources and Irrigation (MoEWRI)

- Formulates national policies, laws, standards and regulations related to energy and water resources.
- Coordinates among government agencies and promotes domestic and foreign investment.
- Plans sustainable and multipurpose utilization of water resources.

b. National Planning Commission (NPC)

- Prepares national development plans and policies.
- Establishes priorities and targets for the energy/hydropower sector.
- Helps allocate government resources and evaluates major development projects.

c. Ministry of Finance (MoF)

- Provides financial policy and budgetary support.
- Coordinates foreign assistance, loans and grants.
- Facilitates government guarantees and financial arrangements for major projects.

ii. **Regulatory and Licensing Institutions**

Department of Electricity Development (DoED) is the principal government department responsible for facilitating and regulating electricity development. It was established to promote the electricity sector and private-sector participation.

Its major roles include:

- Issuing survey and generation licences for hydropower projects.
- Facilitating private-sector participation through a one-window system.
- Coordinating environmental studies such as IEE/EIA.
- Facilitating and regulating hydropower development agreements.
- Monitoring licensed projects and their progress.

iii. **Water Resources Planning and Technical Institutions**

Water and Energy Commission Secretariat (WECS)

- Conducts studies and provides technical advice on water and energy resources.
- Prepares long-term perspectives and plans for integrated water and energy development.
- Helps coordinate the development of water resources among different sectors.

Department of Hydrology and Meteorology (DHM)

- Provides hydrological and meteorological data required for hydropower design.
- Maintains river-flow, rainfall and climate records.
- Supports flood forecasting and assessment of hydrological risks.

Water Resources Research and Development Centre (WRRDC)

- Conducts research and technical studies related to water resources.
- Supports capacity building and development of appropriate technologies.

These organizations are among the major institutions working under or with MoEWRI.

iv. Electricity Generation, Transmission and Distribution Institutions

a. Nepal Electricity Authority (NEA)

NEA is the major public electricity utility of Nepal. Its responsibilities include:

- Planning, constructing and operating power-generation projects.
- Developing and operating transmission systems.
- Distributing electricity to consumers.
- Purchasing electricity from independent power producers (IPPs).
- Selling electricity domestically and participating in cross-border electricity trade.

b. Vidhyut Utpadan Company Limited (VUCL)

- Develops hydropower projects on behalf of the government.
- Undertakes project studies, development and investment in generation projects.

c. Rastriya Prasaran Grid Company Limited (RPGCL)

- Develops and operates major transmission infrastructure.
- Helps establish a strong national transmission grid.
- Facilitates evacuation of electricity generated by hydropower projects.

v. Hydropower Investment and Financing Institutions

Hydroelectricity Investment and Development Company Limited (HIDCL)

- Provides loans and equity investment for hydropower and other energy projects.
- Mobilizes domestic financial resources.
- Supports investment in generation and transmission projects.

Commercial Banks and Financial Institutions

- Provide project financing and working capital.
- Conduct financial appraisal and due diligence.
- Support construction and development of hydropower projects.

International Financial Institutions such as the World Bank, Asian Development Bank and other development partners provide loans, grants, guarantees and technical assistance for suitable projects.

vi. **Private Sector and Independent Power Producers (IPPs)**

Private companies play a major role in Nepal's hydropower development.

Their roles include:

- Conducting feasibility and detailed project studies.
- Arranging equity and debt financing.
- Obtaining necessary government approvals and licenses.
- Operating and maintaining completed projects.
- Selling generated electricity under power purchase agreements (PPAs).

The government encourages private investment as an important means of increasing hydropower generation. DoED specifically facilitates private-sector participation in the electricity sector.

vii. **Environmental and Social Institutions**

Ministry of Forests and Environment (MoFE) and its concerned agencies:

- Review and approve environmental studies as required.
- Ensure compliance with environmental laws.
- Monitor impacts on forests, biodiversity and ecosystems.
- Address environmental mitigation and social issues.

Local Governments (Municipalities and Rural Municipalities)

- Coordinate local-level development activities.
- Facilitate local infrastructure and community coordination.
- Support implementation of local development and social programs associated with projects.

5. **A hydropower plant is to be installed in a river of which the firm flow is 100 cumecs. The net head is 110m and the plant's overall efficiency is 85%. Environmental Regulation restricts that 10% of the flow cannot be divided as it has to be released downstream. What should be the installed capacity of power plant, if it is to be operated for daily peaking for six hours. Assume that 10% of the stored water in regulating pond will be lost in evaporation.**

Answer:

$$\text{Firm flow, } Q_f = 100 \text{ m}^3/\text{s}$$

$$\text{Net head, } H = 110 \text{ m}$$

$$\text{Overall efficiency, } \eta = 85\% = 0.85$$

$$\text{Environmental flow release} = 10\%$$

$$\text{Peaking operation} = 6 \text{ hours/day}$$

$$\text{Evaporation loss in regulating pond} = 10\%$$

Step 1: Calculate flow available for power generation

Since 10% of the river flow must be released downstream:

$$Q_{\text{available}} = 100(1 - 0.10)$$
$$Q_{\text{available}} = 90 \text{ m}^3/\text{s}$$

Step 2: Calculate water available for one day

The plant collects water for 24 hours but operates at peak capacity for only 6 hours.

Therefore, the total daily available water is:

$$V = 90 \times 24 \times 3600 \quad V = 90 \times 24 \times 3600$$

Considering 10 % evaporation loss :

$$V_{\text{usable}} = 0.9 (90 \times 24 \times 3600)$$

Step 3: Calculate design discharge during 6-hour peaking

$$Q_p = \frac{V_{\text{usable}}}{6 \times 3600}$$
$$Q_p = \frac{0.9 (90 \times 24 \times 3600)}{6 \times 3600}$$

$$Q_p = 324 \text{ m}^3/\text{s}$$

Step 4: Calculate installed capacity

The hydropower equation is:

$$P = \rho g Q H \eta$$

Therefore :

$$P = 1000 \times 9.81 \times 324 \times 110 \times 0.85$$
$$P = 297,184,140 \text{ W}$$

Thus:

$$P \approx 297.2 \text{ MW}$$

6. **What are Return of Equity (ROE) and internal rate of return (IRR)? What is the main factor which generally makes ROE more than IRR in the context of financial analysis of a hydropower project.**

Answer:

Return on Equity (ROE)

Return on Equity (ROE) measures the return earned by the equity investors on the money they have invested in a project or company.

$$ROE = \frac{\text{Net Profit After Tax}}{\text{Shareholders' Equity}} \times 100 \%$$

In a hydropower project, it indicates how efficiently the equity investment of the project promoters/shareholders generates profit.

Higher ROE generally indicates a higher return to equity investors.

Internal Rate of Return (IRR)

The Internal Rate of Return (IRR) is the discount rate at which the Net Present Value (NPV) of all project cash flows becomes zero.

$$NPV = \sum_{t=0}^n \frac{C F_t}{(1 + IRR)^t} = 0$$

where:

CF_t = net cash flow in year t

t = time period

n = project life

IRR represents the overall profitability or rate of return of the project considering all investment and cash flows over its lifetime.

The main factor that marks ROE more than IRR in the context of financial analysis of a hydropower project is financial leverage (use of debt financing).

Hydropower projects are usually financed using a combination of: Equity and Debt

If the project earns a return on its total investment that is greater than the effective cost of debt, the use of borrowed money increases the return received by equity investors.

Suppose:

- Total project cost = 100%
- Equity investment = 30%
- Debt financing = 70%

The equity investors invest only 30% of the total capital but receive the remaining profits after paying interest and repaying debt obligations. Therefore, the return on their smaller equity investment can be relatively high.

This is known as financial leverage.

7. **What exactly are Reynolds Number, Froude Number and Mach Number? Explain their significance with reference to their different values.**

Answer:

These are dimensionless numbers used in fluid mechanics to describe and compare different characteristics of fluid flow.

i. Reynolds Number (Re)

The Reynolds number is the ratio of inertial force to viscous force.

$$\mathfrak{R} = \frac{\text{Inertial Force}}{\text{Viscous Force}}$$

It is expressed as:

$$\mathfrak{R} = \frac{\rho VL}{\mu} = \frac{VL}{\nu}$$

where:

ρ = density of fluid

V = velocity of flow

L = characteristic length or diameter

μ = dynamic viscosity

ν = kinematic viscosity

Significance of different Reynolds number values

For flow through a circular pipe:

Reynolds Number	Type of Flow
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Re < 2000	Laminar flow
2000 < Re < 4000	Transitional flow
Re > 4000	Turbulent flow

Low Re: Viscous forces dominate, resulting in smooth and orderly flow.

High Re: Inertial forces dominate, resulting in turbulent and irregular flow.

Significance: Reynolds number is used to determine the nature of fluid flow and is important in pipe, penstock, and turbine hydraulic design.

ii. Froude Number (Fr)

The Froude number is the ratio of inertial force to gravitational force.

$$Fr = \frac{\text{Inertial Force}}{\text{Gravitational Force}}$$

For open-channel flow:

$$Fr = \frac{V}{\sqrt{gL}}$$

For a rectangular open channel:

$$Fr = \frac{V}{\sqrt{gD}}$$

where :

V = mean velocity of flow

g = acceleration due to gravity

L = characteristic length

D = hydraulic depth

Significance of different Froude number values

Froude Number	Type of Flow
Fr<1	Subcritical flow
Fr=1	Critical flow
Fr>1	Supercritical flow

Fr<1: Gravity forces dominate. Surface disturbances can travel upstream.

Fr=1: Flow is at the critical condition.

Fr>1: Inertial forces dominate. Surface disturbances cannot travel upstream.

Significance: Froude number is important in the design of spillways, canals, rivers, hydraulic jumps, and other open-channel structures.

iii. Mach Number (M)

The Mach number is the ratio of the velocity of flow to the speed of sound in the medium.

$$M = \frac{V}{a}$$

where:

V = velocity of the object or fluid

a = speed of sound in the medium

Significance of different Mach number values

Mach Number	Flow Condition
$M < 1$	Subsonic flow
$M = 1$	Sonic flow
$1 < M < 5$	Supersonic flow
$M > 5$	Hypersonic flow

$M < 1$: Compressibility effects are generally small at sufficiently low Mach numbers.

$M = 1$: The flow velocity equals the speed of sound.

$M > 1$: Compressibility becomes highly significant and shock waves may occur.

Significance: Mach number is mainly used in the study and design of compressible fluid flow, such as in aircraft, rockets, jet engines, and high-speed gas flow.

8. **Water is flowing down through a vertical pipe of uniform cross section or cross section of height H . Leaving all other theoretical and practical constraints inside, where would you get the maximum power. If a turbine were installed i) Near the top of the pipe ii) At the mid height of the pipe iii) Near the top end of the pipe iv) Location of turbine does not make any difference? You need to justify your answer quantitatively.**

Answer:

The location of the turbine does not make any difference in the maximum power from a turbine installed in vertical pipe, provided the pipe has a uniform cross-section and all other theoretical and practical losses are neglected.

Consider a vertical pipe of height H , with water flowing downward. Let:

z = vertical distance of the turbine below the top of the pipe.

p_1 = pressure just upstream of the turbine.

p_2 = pressure just downstream of the turbine.

V = velocity of water.

H_T = head extracted by the turbine.

Since the pipe has a uniform cross-section:

$$V_1 = V_2$$

Therefore, the velocity heads on both sides of the turbine are equal.

Apply the Bernoulli equation between a point just above and a point just below the turbine:

$$\frac{p_1}{\rho g} + \frac{V^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V^2}{2g} + z_2 + H_T$$

Since the two points are immediately on opposite sides of the turbine, their elevation difference is negligible:

$$z_1 \approx z_2$$

Thus:

$$H_T = \frac{p_1 - p_2}{\rho g}$$

The available energy at any location is redistributed between pressure head and elevation head.

At the top of the pipe

Near the top:

- Elevation head is maximum.
- Pressure head is relatively low.

At the middle of the pipe

At mid-height:

- Some elevation head has been converted into pressure head.
- The total energy remains the same.

Near the bottom of the pipe

Near the lower end:

- Elevation head is low.
- Pressure head is maximum.

In all cases, neglecting losses, the total available head remains:

$$H_T = H$$

Therefore, the maximum turbine power is:

$$P_{max} = \rho g Q H$$

Considering turbine efficiency:

$$P_{max} = \eta \rho g Q H$$

where :

ρ = density of water

g = acceleration due to gravity

Q = discharge

H = total vertical height of the pipe

η = turbine efficiency

9. Starting from the definition, derive a relationship for ‘specific speed’ as a function of speed power and head. What is the significance and use of specific speed?

Answer:

Specific speed (N_s) of a turbine is defined as the speed of a geometrically similar turbine that would produce unit power under unit head. It is an important parameter used to compare turbines of different sizes and to select a suitable turbine for a given head and power.

Derivation of the Specific Speed Relationship

For geometrically similar turbines operating under similar conditions:

Step 1: Relationship between speed and head

The peripheral velocity of a turbine runner is proportional to the square root of the head:

$$u \propto \sqrt{H}$$

Since:

$$u \propto ND$$

where:

N = rotational speed

D = runner diameter

Therefore:

$$ND \propto \sqrt{H}$$

$\therefore$,

$$N \propto \frac{\sqrt{H}}{D}$$

Step 2: Relationship between discharge and head

For similar turbines:

$$Q \propto D^2 \sqrt{H}$$

Step 3: Relationship between power and head

The power produced by a turbine is:

$$P = \rho g Q H \eta$$

Assuming constant efficiency:

$$P \propto QH$$

Substituting:

$$Q \propto D^2 \sqrt{H}$$

we get:

$$P \propto D^2 H^{3/2}$$

Therefore:

$$D^2 \propto \frac{P}{H^{3/2}}$$

Taking square root:

$$D \propto \frac{\sqrt{P}}{H^{3/4}}$$

Step 4: Substitute D into the speed relationship

We have:

$$N \propto \frac{\sqrt{H}}{D}$$

Substituting:

$$D \propto \frac{\sqrt{P}}{H^{3/4}}$$

Therefore:

$$N \propto \frac{\sqrt{H}}{\sqrt{P} / H^{3/4}}$$

Thus:

$$N \propto \frac{H^{5/4}}{\sqrt{P}}$$

Rearranging:

$$N \sqrt{P} \propto H^{5/4}$$

Therefore, the specific speed of a turbine is:

$$N_s = \frac{N \sqrt{P}}{H^{5/4}}$$

where:

N_s = specific speed

N = rotational speed of turbine (rpm)

P = power output

H = net head

Significance and Uses of Specific Speed:

Specific speed is an important parameter that indicates the operating characteristics of a hydraulic turbine. Its main significance and uses are:

- i. **Selection of turbine type:**
It helps select the most suitable turbine for the available head and power.
 - a. Low specific speed → Pelton turbine
 - b. Medium specific speed → Francis turbine
 - c. High specific speed → Kaplan/Propeller turbine
- ii. **Comparison of turbines:**
It allows turbines of different sizes and capacities to be compared on a common basis.
- iii. **Preliminary turbine design:**
It helps estimate the suitable runner diameter, rotational speed, and general dimensions of a turbine.
- iv. **Determination of operating characteristics:**
It indicates the relationship between the turbine's speed, power, and operating head.
- v. **Selection of generator speed:**
It helps in selecting an appropriate turbine speed that can be suitably coupled with an electrical generator.
- vi. **Assessment of turbine performance:**
Turbines with similar specific speeds generally have similar hydraulic and performance characteristics.

10. **What are the sources of loss of energy stored in hydropower and how is it measured? The supply to a pressure turbine is under a head of 20 m. As the turbine draws $0.5 \text{ m}^3/\text{s}$ of water the head loss in the 300 mm diameter supply line is 2.5m. Calculate the pressure intensity at the entrance to the turbine. Determine the energy absorbed by the turbine in kW. Ignoring the functional losses between the entrance and end of turbine, when a negative pressure $3 \times 10^4 \text{ pa}$ exists at the 500 mm diameter section of the draft tube 1.4 m below the supply line. Also find the output of the turbine, if its efficiency is 85%.**

Answer:

In a hydropower project, the potential energy of water is converted into mechanical energy by the turbine and then into electrical energy by the generator. During this process, part of the available energy is lost due to different hydraulic and mechanical losses.

i. Head Loss in Water Conveyance System

Energy is lost due to friction between flowing water and the surfaces of canals, tunnels, pipes, and penstocks. Major losses include:

- Friction loss in tunnels and pipes
- Loss at bends and curves
- Loss at valves and gates
- Loss at expansion and contraction sections
- Loss at intake and trash racks

The head loss due to friction can be calculated by the Darcy–Weisbach equation:

$$h_f = f \frac{L}{D} \frac{V^2}{2g}$$

ii. Turbine Losses

Some hydraulic energy is lost inside the turbine because of:

- Hydraulic friction
- Shock losses

- Leakage
- Mechanical friction

These losses reduce the amount of water energy converted into useful mechanical energy.

iii. Generator Losses

Mechanical energy supplied by the turbine is not completely converted into electrical energy. Losses occur due to:

- Copper losses in windings
- Iron or core losses
- Mechanical losses due to friction and ventilation

iv. Transformer and Transmission Losses

After generation, electrical energy may be lost through:

- Transformer losses
- Resistance in transmission lines
- Switching and auxiliary equipment losses

v. Leakage and Seepage Losses

Water and energy may be lost through:

- Leakage from canals and tunnels
- Seepage through reservoirs
- Leakage from pipes and penstock

Measurement of Energy Losses

The energy loss in a hydropower system is generally measured in terms of head loss or by comparing input and output power.

a. By measuring head loss

The difference between the gross head and net head represents the hydraulic energy lost during water conveyance:

$$H_{loss} = H_{gross} - H_{net}$$

The hydraulic power loss is:

$$P_{loss} = \rho g Q H_{loss}$$

where :

ρ = density of water

g = acceleration due to gravity

Q = discharge

H_{loss} = headloss

b. By measuring efficiency

Overall efficiency of the hydropower plant is:

$$\eta = \frac{\text{Electrical Power Output}}{\text{Hydraulic Power Input}} \times 100\%$$

Thus, the energy loss can be determined from:

$$\text{Energy Loss} = \text{Input Energy} - \text{Output Energy}$$

11. What is field investigation for a hydropower project? Differentiate between geological and geotechnical investigations. Explain the step-by-step procedure to estimate the probabilistic seismic intensity for target probability of exceedance in specified years at a hydropower project site.

Answer:

Field investigation is the systematic collection and examination of physical, geological, hydrological, environmental, and socio-economic information at the proposed hydropower project site. It is carried out through site visits, surveys, measurements, mapping, sampling, and testing to determine the feasibility and suitability of the project.

The main purposes of field investigation are:

- i. **Topographical investigation:** to study the terrain, elevation, river alignment, and suitable locations for project components.
- ii. **Hydrological investigation:** to assess river discharge, flow variation, flood levels, sediment load, and water availability.
- iii. **Geological and geotechnical investigation:** to examine rock and soil conditions, faults, landslides, slope stability, and foundation conditions.
- iv. **Site investigation for structures:** to select suitable locations for the dam/diversion weir, intake, tunnel, powerhouse, and transmission facilities.
- v. **Environmental investigation:** to identify possible impacts on forests, wildlife, water quality, and the surrounding ecosystem.
- vi. **Socio-economic investigation:** to study settlements, land acquisition, affected people, local infrastructure, and possible social impacts.
- vii. **Access and construction material investigation:** to identify access roads, availability of construction materials, and suitable disposal or quarry sites.

Both geological and geotechnical investigations are essential during the field investigation of a hydropower project. However, geological investigation mainly studies the natural geological conditions of the project area, whereas geotechnical investigation determines the engineering properties of soil and rock for design and construction.

Geological Investigation	Geotechnical Investigation
The main purpose of geological investigation is to understand the geological conditions and history of the project area.	The main purpose of geotechnical investigation is to determine engineering properties of soil and rock.
Main focus of geological investigation are rock types, geological structures, faults, folds, joints, and landslides	Main focus of geotechnical investigation is strength, deformation, permeability, bearing capacity, and other engineering properties
The nature of study is mainly qualitative and regional/site geological assessment.	The nature of study is mainly quantitative engineering assessment.
Geological investigation includes geological mapping, surface observation, identification of faults and discontinuities	Geotechnical investigation includes borehole drilling, test pits, sampling, in-situ and laboratory testing
Rock classification also identifies rock type and geological formation	Geotechnical investigation determines rock mass quality and engineering behavior
Limited engineering testing which focuses mainly on geological interpretation	Extensive laboratory and field tests are conducted in geotechnical investigation.
Selection of project site and identification of geological hazards	Detailed design of foundations and underground structures
It helps select suitable locations for dam, tunnel, powerhouse, and other structures	It helps design dam foundations, tunnel support, slopes, and underground excavations

Probabilistic Seismic Hazard Analysis is used to estimate the seismic intensity expected at a hydropower project site for a specified probability of exceedance within a given time period. Seismic intensity may be expressed as Peak Ground Acceleration (PGA) or spectral acceleration.

i. **Define the project site and target seismic hazard:**

Identify the exact location of the proposed hydropower project and define site coordinates, required design life or exposure period, target probability of exceedance, P etc.

For example: Estimate the PGA having a 10% probability of exceedance in 50 years.

ii. **Collect regional seismic and geological data**

Collect all available information on earthquakes and seismic sources around the project site, including historical earthquake records, instrumental earthquake catalogue, magnitude of past earthquakes etc.

iii. **Prepare the earthquake catalogue**

Develop a reliable earthquake database by:

1. Combining historical and instrumental earthquake records.
2. Removing duplicate events.
3. Removing foreshocks and aftershocks (declustering).
4. Checking catalogue completeness.
5. Converting earthquake magnitudes into a common magnitude scale, usually moment magnitude M_w .

This catalogue is used to determine the seismicity of the region.

iv. **Identify and model seismic sources**

Identify all potential earthquake-generating sources that can influence the project site. These may include active faults, fault zones, subduction zones, diffuse seismic zones

Each source is characterized by location, geometry, distance from the project site, earthquake occurrence rate etc.

v. **Develop the earthquake recurrence relationship**

The frequency of earthquakes of different magnitudes is estimated using the Gutenberg–Richter relationship:

$$\log_{10}N(M)=a-bM$$

where:

- $N(M)$ = number of earthquakes with magnitude greater than or equal to M
- a and b = seismicity constants

This relationship provides the probability or annual rate of occurrence of earthquakes of different magnitudes.

vi. **Determine the source-to-site distance**

Calculate the distance between each potential seismic source and the hydropower project site. Different types of distances may be used, such as epicentral distance, hypo central distance etc.

The selected distance measure should be consistent with the Ground Motion Prediction Equation being used.

vii. **Select appropriate Ground Motion Prediction Equations (GMPEs)**

Select suitable GMPEs to estimate ground motion at the site based on:

- Earthquake magnitude
- Source-to-site distance
- Fault mechanism
- Local site conditions

The GMPE estimates the probability distribution of ground-motion intensity such as:

PGA

Or

$S_a(T)$

where $S_a(T)$ is spectral acceleration at a particular structural period.

viii. **Consider local site conditions**

Investigate the geological and geotechnical conditions at the project site, including rock and soil type, shear-wave velocity (V_{s30}), depth of soil deposits etc.

The ground motion calculated for bedrock may be amplified or modified by local soil conditions.

ix. **Calculate the annual probability or rate of exceedance**

For every possible earthquake magnitude and distance, calculate the probability that a selected ground-motion level will be exceeded.

Conceptually:

$$\lambda(IM > x)$$

represents the annual frequency at which an intensity measure IM exceeds a value x.

The result is called the seismic hazard curve.

x. **Develop the seismic hazard curve**

Plot:

- Ground-motion intensity (PGA or spectral acceleration) on the horizontal axis.
- Annual probability/rate of exceedance on the vertical axis.

The curve shows the likelihood of exceeding different levels of seismic intensity at the hydropower project site.

xi. **Convert the target probability of exceedance into annual probability**

For a target probability of exceedance PP in TT years, assuming a Poisson earthquake occurrence model:

$$P = 1 - e^{-\lambda T}$$

Therefore, the annual rate of exceedance is:

$$\lambda = -\ln(1-P)/T$$

where:

- P = target probability of exceedance
- T = specified number of years
- λ = annual rate of exceedance

xii. **Determine the corresponding seismic intensity**

Using the calculated value of λ , enter the seismic hazard curve and determine the corresponding:

- Peak Ground Acceleration (PGA), or
- Spectral acceleration

This is the probabilistic seismic intensity corresponding to the target probability of exceedance.

Example

Suppose the design requirement is: 10% probability of exceedance in 50 years.

Then:

$$\begin{aligned}\lambda &= -\ln(1-0.10)/50 \\ \lambda &= 0.002107 \text{ per year}\end{aligned}$$

The equivalent return period is approximately:

$$T_R = 1/\lambda \approx 475 \text{ years}$$

Therefore, the seismic hazard curve is used to determine the PGA corresponding approximately to a 475-year return period.